

Examen final : Compléments d'informatique

ENSAI 2A UE3

2025-09-29

Instructions

- Written exam
- Duration: 2 hours
- One A4 sheet of **handwritten** notes (front and back) allowed
- No calculator

! Read Before Starting

- You can answer the exercises in any order.
- Within each exercise, the questions are not necessarily ordered by increasing difficulty.
- Unless otherwise stated, the questions expect short answers.
- Strictly follow the instructions. If a question seems unclear, state your assumptions on your answer sheet.
- When code is required, follow the basic language conventions (keywords, indentation, etc.). Syntax errors due to handwriting will be tolerated. If you do not remember the exact name of a Python method, you may use pseudocode.
- Unless explicitly requested, code documentation is not required.

1 Course Questions (7 points)

i Instructions

Give short answers to the following questions.

- a. List the main tools and software used during the practical sessions.
- b. Why should we write quality code?
- c. What are the benefits of dividing an application into layers?
- d. How would you define the relationships between the following classes?
 - Bicycle
 - TwoWheeledVehicle
 - Scooter
 - ElectricBicycle
- e. What is a design pattern?
- f. What principles should a good unit test follow?
- g. What is the purpose of a mock?
- h. Identify the errors and bad practices in the following code. What consequences could they have?

```
from dao.db_connection import DBConnection
from business_object.player import Player

class PlayerDao(metaclass=Singleton):
    def connect(self, pseudo, pwd) -> Player:
        with DBConnection().connection as connection:
            with connection.cursor() as cursor:
                query = " SELECT *                                "
                query += " FROM player                            "
                query += " WHERE pseudo = '" + pseudo + "'"      "
                query += " AND pwd = '" + pwd + "';              "

                cursor.execute(query)
                res = cursor.fetchall()

            return res
```

2 Webservices (3 points)

Instructions

Give short answers to the following questions.

a. Name the different elements of the following request:

`GET https://my-awesome-ws.fr/attack/8`

b. In your opinion, what could the following request do?

`PUT https://my-awesome-ws.fr/pokemon/15`

c. What needs to be added for this request to work?

d. Which HTTP methods correspond to the CRUD operations?

3 Code (6 points)

i Instructions

Respect the principles of layered programming throughout this exercise.

Create a `Person` class with the following attributes:

- last name
 - first name
 - city
 - age
 - is_adult
- a. Create the class and its constructor. Each new person has an age of 0 and is not an adult.
 - b. Add a `birthday()` method that increments the person's age. When the person reaches 18 years old, they become an adult.

Write the documentation for this method.

- c. Using `pytest`, write the unit tests for the `birthday()` method.
- d. Assuming there is a database table `person` containing the same information as the `Person` class, write the SQL query that displays the number of people grouped by city.
- e. The `PersonDao` class provides a `list_all()` method returning all people.

Write a service method that returns only the adults.

- f. **Bonus:** Modify the service method so that it accepts an optional parameter `cities` (a list of cities).
 - If the parameter is provided, return only adults whose city belongs to this list.
 - Otherwise, return all adults.

4 Git and Version Control (4 points)

Your project manager asks you to update the file `src/main.py` in the Git repository:

<https://github.com/soleil59/sustainable-cultures.git>

The update consists of connecting to a database to retrieve the contents of a table named `chicory`.

You have never worked on this repository before, but you have the required permissions and a valid access token.

a. How do you retrieve the repository onto your local machine?

You then modify the `src/main.py` file.

Following good development and security practices, you do **not** hardcode the database credentials. Instead, you store them in a new `.env` file.

b. How do you send your modifications to the remote repository?

You followed the procedure correctly but encounter a merge conflict.

c. After which Git command can such a conflict occur? Briefly explain.

d. What could have caused this conflict?

e. How do you resolve it?